

WavetableController Rev 3 status: open items only

Trey's 9 October files measured against the 7 October Rev 2 Design Review. Done items removed. Power is 9 V from the barrel jack. The actual LED part is unknown to this review: the schematic names a Cree CLS6B-FKW, the footprint is for a different package, and the BOM has no LED line, so polarity statements are relative to the symbol's pin names. The 7 Oct count of 225 ground stitching vias does not hold: the 9 Oct board has 244 vias, none on GND. Generated 2026-10-09.

"Fix before ordering" items

#	Asked for (7 Oct)	Rev 3 (9 Oct) status	Evidence	Fix	Potential consequence if left
2c	Feed U6 from the C19 node, before R31	NOT DONE	FB7 still hangs on /+9V_FLAG after R31/R32; U6's 0.2–0.4 A still flows through them	Move FB7's input from /+9V_FLAG to Net-(C19-Pad1); R31/R32 then carry only LED current	/+9V_FLAG sags to ~5.8 V at 9 V in and moves with display content; the LED rail (and LED brightness) follows what the screen is showing. Not damaging now that the resistors are 1 W

3b	Turn each LED around (9 V → R → anode → cathode → switch)	NOT DONE in the files — covered by a documented 180° hand placement	anodes (Cree pads 1/3/5) on the collectors, cathodes via 300 Ω to +9V_FLAG — still reverse-biased, still 5.5–8.7 V reverse vs 5 V (R3-P2-01)	The actual LED part is unknown to this review: the schematic names a Cree CLS6B-FKW, the footprint is for a different package, and the BOM has no LED line, so every polarity statement is relative to the symbol's pin names, not a verified part. Hand-build workaround (agreed 9 Oct): Trey mounts each LED rotated 180° from the silk pin-1 mark. On this three-row footprint that puts every anode on the 300 Ω side and every cathode on the collector; red and blue swap channels (two-line firmware change). Works only for a two-column 5050 whose colour pairs sit across a row; cannot work for the Cree CLS6B-FKW named on the schematic (single-row part) or for a 5050 with diagonal pairs. Fix the files when convenient: flip the six dice in the schematic and re-annotate the PCB so silk, symbol and netlist match how the part is mounted	Built as the files say: LEDs never light, each die holds 5.5 V reverse against a 5 V rating. Built with the 180° placement: they work. The silk, symbol and netlist then say the opposite of how the part is mounted, so anyone building from the files later gets dark LEDs again
3c	300 Ω resistors → 1206	NOT DONE	R21, R24–R28 still 0603 (R3-P2-04)	Change R21, R24–R28 to 1206 (250 mW); or re-value for the real ~5.8 V rail	With the 9 V adapter the 0603s dissipate 22–43 mW, within rating; the only consequence is dim LEDs (8–12 mA instead of 20 mA, because the rail behind R31/R32 is ≈5.8 V, not 9 V). The 1206 size only matters if a 12 V adapter, the jack's ceiling, is ever used

3d	(new) LED footprint	NEW PROBLEM	actual LED part unknown; custom 2-column 5050 footprint does not fit CLS6B-FKW 4.7 × 1.5 mm (R3-P2-02); LEDs still not on BOM (R3-P4-01)	Decide the actual LED; use its footprint (stock LED_Cree-PLCC6_4.7x1.5mm for CLS6B-FKW); add the BOM line and set the symbol pin names to the real pinout	Either the named part cannot be placed on the pads, or a generic 5050 goes on with an unverified pinout and the polarity check is wrong again. Stencil has 12 LED apertures the assembler cannot fill
5	Move J8 to y 86.3, add four M2 holes at (135.0/209.2, 84.3/126.8), module outline in footprint	NOT DONE	J8 pin 1 still at (196.26, 81.30); 0 mounting holes; projected hole sites collide with R2/R5/U1 (R3-P6-05/06)	Move J8 down 5 mm, add the four 2.4 mm holes and standoffs, move U1/R2/R5 tracks clear, put the outline and holes in the J8 footprint	Display hangs on 20 solder joints; pressing the glass through the panel window loads them and cracks joints or the module PCB over time. Enclosure window has to be cut before the hole positions are final
6b	L1 on the 2016 (DFE201610P) footprint	NOT DONE	still L_0805_2012Metric (R3-P4-02)	Replace the footprint with stock Inductor_SMD:L_Murata_DF E201610P	Pads 0.4 mm narrower than the terminations, ~50 % overlap, 0.5 mm overhang: a weak joint on the one part carrying the switching current; intermittent display rail or tombstoning at reflow
7	Encoder A/B: 10 k pull-up + 10 nF per line	NOT DONE	/ENCL_A, /ENCL_B, /ENCR_A, /ENCR_B are 2-node nets (encoder + Daisy only)	Add 4 × 10 k to 3V3_D and 4 × 10 nF to GND near the Daisy on pins 12, 13, 32, 33; use a state-table decoder in firmware	Skipped or doubled detents on fast or heavy-detent turns, worse with 80–146 mm of unterminated trace on internal pull-ups; looks like a firmware bug
9	Delete D6; bridge to SMA/SOD-123FL	NOT DONE	D6 present (C727114 SOD-323); D2–D5 unchanged (R3-P2-06)	Remove D6 (link its pads); change D2–D5 to an SS14-class SMA/SOD-123FL part	At 0.49 A each conducting diode runs ~0.2 W in a 250 mW SOD-323; junction near 125 °C in a warm box, shortened life or a failed bridge. D6 costs 0.45 V and 0.2 W for no protection the bridge doesn't already give
—	SW1 contact code	unverifiable here	not on the JLC BOM	Check the ordered part is 1101A4VQEA with silver 5 A contacts, not the gold 0.4 VA logic-level variant	A logic-level contact at 0.5 A pits and goes intermittent; power drop-outs on the switch

Fab margins

Item	7 Oct ask	Rev 3	Fix	Potential consequence if left
MK1 annular ring 0.175 → 1.1 mm pad	NOT DONE	0.175 mm (R3-P1-02)	Set MK1 pads 1/2 to 1.1 mm diameter	Below JLC's 0.18 absolute minimum: a stricter lot rejects the order, or a thin ring breaks out and the mic pin lifts
P2 NPTH-to-GND-pad 0.197 → shrink pads 0.05	NOT DONE	0.197 mm (R3-P1-03)	Shrink P2 A1/A12/B1/B12 by 0.05 mm, or nudge them away from the NPTH	3 µm under the 0.20 limit: fabbed OK twice, but a DFM rejection or a drill break-out into the GND pad is possible
U6 thermal vias → 0.3 hole / 0.7 pad	NOT DONE	0.25 hole / 0.6 pad (R3-P1-04)	Edit the U6 footprint's two EP vias to 0.3/0.7	Extra 0.25 mm drill tool; ring 0.175 under the 0.18 minimum; trips the project's own DRC so real errors hide among known ones
Silk outlines 0.12 → 0.15 mm	NOT DONE	0.12 mm (R3-P6-02)	Set footprint outline width to 0.15	Outlines print faint or are dropped; cosmetic
DRC min track/clearance 0 → 0.1	NOT DONE	still 0.0 (R3-P1-14)	Board Setup → Constraints: min track 0.1, min clearance 0.1	KiCad never warns about sub-minimum copper; the next 0.1 mm or thinner trace ships unnoticed

"Worth tidying"

Item	Rev 3	Fix	Potential consequence if left
100 nF + 10 µF at the SD socket P1	NOT DONE — still only C1, 120 mm away (R3-P2-10)	Add 100 nF + 10 µF across P1 VDD/VSS; 10 µF at A1.38	Rail dips at the card on 100 mA write bursts: CRC errors, fallback to 1-bit, the likeliest cause of the Round 1 4-bit failure. MIDI opto shares the dip
TPA6110A2 10 µF bulk + 5 pF across R14/R15	NOT DONE (R3-P2-11)	10 µF (or more) on 3V3_A at U3; 5 pF across R14 and R15	Distortion or a faint whine with long headphone leads; mic amp on the same under-filtered rail
AUDIO_IN_L beside USART1_RX; AUDIO_OUT_R beside TAC_SWITCH_2	NOT DONE — UART still 49 mm at 0.41 mm from AUDIO_IN_L; now the closest pairing on the board (R3-P5-04)	Move the UART tracks ≥0.5 mm (ideally 3× width) from AUDIO_IN_L, or put them on the other layer	A tick on the audio input on MIDI traffic; a click on a button press
Firmware: SPI1 transmit-only (MISO = TAC_SHIFT_R)	firmware, unchanged (pin 10 still TAC_SHIFT_R)	Open SPI1 as TX-only in libDaisy	Right shift button dies the moment the display driver initialises; looks like a hardware fault
ERC: 14 NC flags + 4 PWR_FLAGS	NOT DONE (R3-P3-06)	Add NC flags on P2 SS/SBU pins and J8 3/9/15/18; PWR_FLAG on /DSY_VIN, Net-(U6-EN), Net-(U4-VDD), GND	A real wiring error hides among 20 known false errors on the next revision

Chassis standoffs near the jack row	NOT DONE — 0 mounting holes (R3-P6-05)	Add ≥ 2 standoff holes within ~ 15 mm of J3–J7; close-fit panel holes	Every plug insertion flexes the board about the encoder nuts; cracked jack joints over time
Hand-assembly BOM sheet	NOT DONE (R3-P4-03)	Second BOM sheet listing the 18 THT parts incl. C23, C24, J8, Headphones1	A board passes assembly and fails on the bench because a through-hole cap at the display was never fitted

New in the 9 October files

Item	What was measured	Fix	Potential consequence if left
No ground stitching vias	244 vias, 0 on GND; the front pour is 15 fragments joined to the back plane only through component holes; 3 fragments hang on a single pad; 10 bottom-side GND pads on U2/U3/U4/U6 reach ground through one thin track	Add stitching vias every 10–15 mm and at every front-pour fragment and IC ground	Long ground returns under the audio and SPI runs: higher noise floor, weaker ground for the buck; single-pad fragments go floating if that joint cracks
Eight single-spoke thermal reliefs	GND pads on TAC_SWITCH_1, U1, P2 (two), R18, C9, U3, C10 connect through one 0.5 mm spoke	Reduce the zone thermal gap or add a short GND track to each	Works today; one cracked spoke opens a ground pin
No inrush limit	200 μF (C15 + C19) behind only the beads and Schottkys; 700 μF total	A few ohms or an NTC ahead of C15, or fewer bulk caps	About 26 A peak at plug-in or when SW1 is flipped, through diodes rated 9 A surge and 2 A beads; contact wear on SW1 and the barrel jack
Display reset line floats	RES_SPI is a two-node net with no pull-up or RC	10 k from RES_SPI to +3V3_OLED	Undefined reset state until firmware drives it; a glitchy display at power-up
Headphone pot is a linear taper	Headphones1 is B10k, signal on pin 1, GND on pin 3	A-taper part; confirm clockwise is louder with the PTD902 terminal order	Volume bunched at one end of the travel; possibly backwards
Pot bushing outside the outline	Bushing exits 9 mm past the bottom edge, 16.75 mm from the headphone jack	Panel cut-out, or move the pot inboard	Panel does not fit without a cut-out
"Exclude from position files" inconsistent	Set on J3–J7 only; 13 other through-hole parts and the LEDs are in the pick-and-place file	Set the flag on every hand-soldered part	15 no-match entries at JLCPCB upload bury the two that matter, the LEDs
Reference designators repurposed since August	C14 was the 470 μF noise-fix cap, now 100 nF; C17 100 nF $\rightarrow$ 100 μF ; C15 10 μF $\rightarrow$ 100 μF	A line in the change log	Old notes, rework instructions and photos naming C14, C15 or C17 point at the wrong part